

Dynamic Margin CLAP: Adaptive Positive-Pair Margins for Audio-Text Retrieval

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Abstract

Contrastive audio-language pre-training typically applies uniform optimisation pressure to all matched audio-caption pairs, despite substantial differences in caption specificity and semantic ambiguity. We propose **Dynamic Margin CLAP**, a training-time intervention that assigns each positive pair a margin derived from the normalised inverse document frequency of its caption tokens. The margin raises the alignment requirement for semantically specific pairs and relaxes it for generic or ambiguous descriptions, requires no external ontology, and is removed entirely at inference.

We evaluate the proposed margin across multiple audio-caption corpora, Euclidean and Lorentz-hyperbolic embedding spaces, two hyperbolic parameterizations, and a multi-positive retrieval protocol. The margin consistently improves audio-text retrieval regardless of the underlying geometry, whereas the benefit of curvature is narrower, corpus-dependent, and accompanied by additional retrieval cost. Comparisons with long-tail re-weighting show that changing the positive logit inside the softmax is not equivalent to rescaling whole examples. Margin-source ablations further indicate that most of the improvement comes from introducing a positive-pair margin itself, while the IDF-based specificity signal provides a smaller additional benefit. These results position adaptive positive-pair margins as the primary contribution and hyperbolic geometry as a complementary option.

CCS Concepts

• **Computing methodologies** → **Neural networks**; *Unsupervised learning*; • **Information systems** → Similarity measures.

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Keywords

Dynamic Margins, Audio-Text Retrieval, Contrastive Learning, Hyperbolic Geometry

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1 Introduction

Audio understanding has undergone a fundamental shift with the advent of large-scale contrastive pre-training. CLAP [8] established a dominant framework by aligning audio and natural language descriptions in a shared embedding space, enabling zero-shot recognition and cross-modal retrieval without task-specific supervision. Subsequent work has scaled this paradigm along multiple axes: larger and more diverse training corpora [23, 37], improved audio encoders [2], and instruction-tuned text encoders [7], positioning contrastive audio-language models as the de facto backbone for audio captioning [18], sound event detection [11], and audio question answering [7].

Despite this progress, the positive side of the contrastive objective remains largely unchanged: standard InfoNCE [33] applies the same optimisation pressure to every matched pair. This uniform treatment ignores a defining property of audio captions. Some captions identify a specific source, event, or timbral attribute, whereas others provide only a broad scene description; these pairs should not necessarily face the same alignment requirement. Hard-negative mining [29] refines the negative set but leaves this positive-pair asymmetry unresolved, and loss re-weighting changes the contribution of an example without changing its decision boundary inside the softmax.

A related but distinct question concerns representation geometry. Existing CLAP variants inherit the Euclidean space of CLIP [27], although audio concepts span nested acoustic scenes, events, sources, and fine-grained timbral categories with heavy-tailed occurrence frequencies [11, 22]. Lorentz hyperbolic space offers exponentially

growing volume and may represent this branching structure with lower distortion [24, 30]. However, geometry changes representational capacity, while a margin changes optimisation pressure; neither mechanism implies the success of the other.

We propose **Dynamic Margin CLAP**, centred on a per-pair positive margin computed from the normalised inverse document frequency (IDF) of caption tokens. Semantically specific captions receive a larger margin and therefore a harder alignment target, while generic captions receive a smaller one. The statistic is obtained from the active training corpus, needs no external ontology or learned auxiliary model, and is discarded at inference. To test whether the effect depends on embedding curvature, we apply the same margin to Euclidean CLAP and to two Lorentz-hyperbolic parameterizations. Across a reconstructed WavCaps–FreeSound split, Clotho, and MACS, the margin improves every tested geometry; curvature provides an additional benefit only on the largest and most heterogeneous corpus. Our contributions are:

- A dynamic positive-pair margin for audio–language contrastive learning, using corpus-derived caption specificity without an ontology or inference-time component.
- A controlled evaluation across three corpora, three geometry choices, two hyperbolic lifts, and multi-positive retrieval, showing that the margin is stable while curvature is a conditional bias.
- Controls against IDF and focal-style re-weighting, together with margin-source ablations and retrieval-cost analysis, separating the margin effect from the smaller IDF increment and curvature cost.

2 Related Work

Adaptive margins in contrastive learning. Additive margin losses originate in metric learning, from triplet networks [31] to angular margin penalties such as ArcFace [5] and CosFace [34]. In self-supervised contrastive learning [4, 16], uniform temperature is the dominant regulariser, while hard-negative mining [29] changes the negative set but leaves positive-pair pressure uniform. Hierarchical margin schedules [19, 38] have been explored in supervised visual settings, but they rely on external class hierarchies and do not address ontology-free audio–language alignment. Multi-grained audio–language methods likewise operate without per-pair margin modulation [32]. Dynamic Margin CLAP instead derives a positive-pair margin directly from caption statistics and applies it inside the cross-modal softmax.

Hyperbolic representation learning. Nickel and Kiela [24] showed that hyperbolic space embeds hierarchical structures with exponentially lower distortion than Euclidean space. MERU [6] extended this principle to vision–language alignment, and HOVER [35] reported analogous gains for video–text retrieval; [25, 36] develop geometry-grounded and compositional variants, while Ge et al. [10] study geodesic similarity inside contrastive learning. In audio, hyperbolic geometry has been applied to source separation [26], music representation [28], supervised classification [13], and audio–visual zero-shot learning [14]. Chen et al. [3] independently motivate hyperbolic language representations. We treat Lorentz projection as an optional geometry for testing the generality of the proposed margin, rather than assuming curvature is universally beneficial.

Long-tail audio learning. Re-sampling [12] and label-hierarchy regularisation [15] address long-tail imbalance in audio classification, but target discriminative objectives rather than matched audio–caption alignment. A token-derived margin may appear similar to rare-example re-weighting, yet the mechanisms differ: re-weighting rescales a complete loss after the softmax, whereas a margin changes the matched logit and hence the required similarity gap. Section 5.4 compares these alternatives under identical encoders, geometry, and optimisation settings. The results support the narrower claim that positive-pair margins and long-tail re-weighting are not interchangeable; they do not establish which semantic subgroups account for the aggregate improvement (Section 5.9).

3 Preliminaries

3.1 Contrastive Audio–Language Pre-training

CLAP [8] trains a dual-encoder model—audio encoder f_a and text encoder f_t —by minimising a symmetric InfoNCE objective over matched audio–caption pairs:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{e^{\mathbf{a}_i^\top \mathbf{t}_i / \tau}}{\sum_j e^{\mathbf{a}_i^\top \mathbf{t}_j / \tau}} + \log \frac{e^{\mathbf{t}_i^\top \mathbf{a}_i / \tau}}{\sum_j e^{\mathbf{t}_j^\top \mathbf{a}_i / \tau}} \right]. \quad (1)$$

Here $\mathbf{a}_i = f_a(\mathbf{x}_i)$ and $\mathbf{t}_i = f_t(s_i)$ are the ℓ_2 -normalised embeddings of the i -th audio clip $\mathbf{x}_i$ and its caption string s_i , N is the batch size, and τ is the temperature. All existing CLAP variants project both modalities into a shared Euclidean space and score with a dot product, inheriting the geometry of CLIP [27] without asking whether flat geometry suits audio semantics.

3.2 Lorentz Hyperbolic Space

The Lorentz model of n -dimensional hyperbolic space with curvature $c > 0$ is defined as the upper sheet of a two-sheeted hyperboloid embedded in Minkowski space $\mathbb{R}^{n+1}$:

$$\mathbb{L}_c^n = \{\mathbf{x} \in \mathbb{R}^{n+1} : \langle \mathbf{x}, \mathbf{x} \rangle_{\mathcal{L}} = -1/c, x_0 > 0\}, \quad (2)$$

where $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}} = -x_0 y_0 + \sum_{i=1}^n x_i y_i$ is the Lorentzian inner product. The geodesic distance between $\mathbf{x}, \mathbf{y} \in \mathbb{L}_c^n$ is:

$$d_{\mathcal{L}}(\mathbf{x}, \mathbf{y}) = \frac{1}{\sqrt{c}} \cosh^{-1}(-c \langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}}). \quad (3)$$

A key property of $\mathbb{L}_c^n$ is that the volume of a geodesic ball of radius r grows as $O(e^{r\sqrt{c}})$, exponentially faster than Euclidean space, making it well-suited to embedding hierarchical structures with low distortion [24, 30]. We use the exponential map at the origin $\mathbf{o} = (1/\sqrt{c}, \mathbf{0})$:

$$\exp_{\mathbf{o}}(\mathbf{v}) = \cosh(\sqrt{c} \|\mathbf{v}\|) \mathbf{o} + \frac{\sinh(\sqrt{c} \|\mathbf{v}\|)}{\sqrt{c} \|\mathbf{v}\|} \mathbf{v}, \quad (4)$$

where $\mathbf{v} \in T_{\mathbf{o}}\mathbb{L}_c^n$ is a tangent vector at the origin. We adopt the Lorentz model over the Poincaré ball [24] for its superior numerical stability under large-batch gradient optimization [1, 9]. Figure 1 illustrates the induced radial hierarchy: abstract concepts cluster near the origin while semantically specific samples are pushed to larger geodesic radii.

Audio captions motivate both ingredients of our method: they describe events at very different levels of granularity, which calls

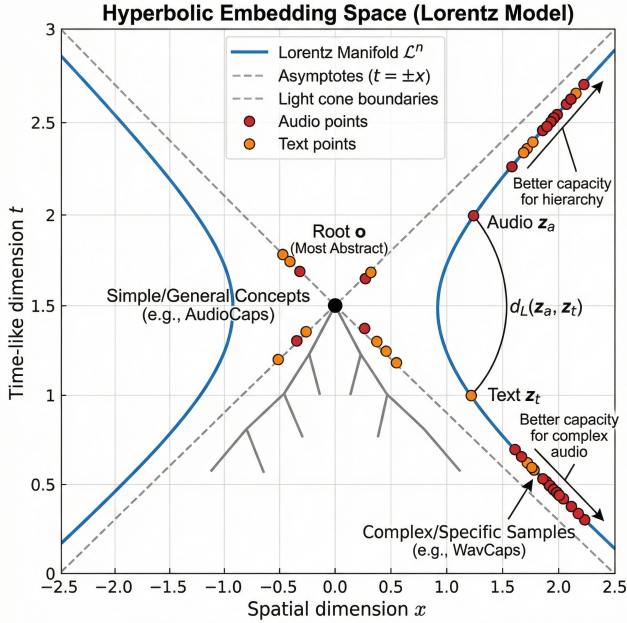


Figure 1: The Lorentz hyperboloid $\mathbb{L}_c^n$ used as embedding space. The origin o acts as an implicit semantic root; general concepts cluster near the origin while specific samples are placed at larger geodesic radii $d_L(z_a, z_t)$. Euclidean and spherical geometries have polynomial volume growth and cannot host the same radial spread at comparable distortion. Corpus names in the figure are illustrative; the evaluation set is given in Table 1.

for volume that grows faster than polynomially with radius, and the events themselves occur with heavy-tailed frequency, which calls for optimisation pressure that is not identical for every positive pair.

3.3 Margin and Geometry as Orthogonal Factors

The positive-pair margin and the embedding geometry intervene at separate points in contrastive learning. Subtracting m_i from a matched logit lowers the score assigned to the correct pair during training and therefore requires a larger positive–negative similarity gap to attain the same objective value. This intervention is local to optimisation: it does not alter the metric, embedding dimension, candidate set, or inference rule. A uniform margin imposes the same additional gap on every pair, whereas the proposed dynamic form varies that gap using a caption-derived specificity score. Consequently, the margin can improve a Euclidean model without a hierarchical manifold or any deployment-time module.

Geometry controls a different quantity: the family of pairwise relations that can be represented with low distortion. Euclidean space allocates volume polynomially with radius, whereas Lorentz space allocates it exponentially and can place many fine-grained descendants around broader semantic regions. This capacity motivates hyperbolic projection, but does not guarantee that optimisation will discover a useful radial organisation on every corpus. A hyperbolic

model can therefore fail to outperform a Euclidean one even when the same margin remains effective.

This decomposition determines our factorial evaluation. We apply an identical margin schedule to Euclidean, DirectLift, and ExpMap models and compare each margin variant with its own no-margin control. If the margin transfers across all three similarities, its benefit is not specific to curvature. Conversely, differences between DirectLift and ExpMap reveal sensitivity in how hyperbolic radii are allocated rather than in the margin itself. We therefore treat adaptive positive-pair pressure as the general intervention and Lorentz geometry as a conditional inductive bias.

4 Method

4.1 Model Architecture

Dynamic Margin CLAP adopts the CLAP dual-encoder framework [8] with frozen HTS-AT [2] and RoBERTa [20] encoders. Freezing both backbones isolates the proposed margin and the optional geometry choice from changes in encoder capacity. The encoders are followed by modality-specific linear projection heads $g_a, g_t : \mathbb{R}^d \rightarrow \mathbb{R}^n$ mapping into the tangent space at the origin of $\mathbb{L}_c^n$, with a residual connection of weight λ before the lift to stabilise early training. Writing x for the raw audio and s for the caption string, the projected representations are placed on the hyperboloid by

$$z_a = \Phi(g_a(f_a(x))), \quad z_t = \Phi(g_t(f_t(s))), \quad (5)$$

giving hyperbolic embeddings $z_a, z_t \in \mathbb{L}_c^n$. We use z_a, z_t for final Lorentz embeddings throughout the paper and in Figures 1 and 2. For the Euclidean control, the same projection heads are ℓ_2 -normalised directly and the dynamic margin is applied to their dot-product logits; the margin itself therefore does not depend on the lift Φ .

Two parameterizations of the lift Φ . There are two natural ways to place a projected vector $v \in \mathbb{R}^n$ on $\mathbb{L}_c^n$, and they are not equivalent in practice, so we evaluate both. **DirectLift**, our default, treats v as the spatial coordinates and recovers the time coordinate from the hyperboloid constraint, $\Phi(v) = (\sqrt{\|v\|^2 + 1/c}, v)$; it is cheaper and better conditioned for large $\|v\|$, but compresses radii. **ExpMap** sets $\Phi = \exp_o$ (Eq. 4), so that $\|v\|$ is the geodesic radius of the result. The two induce different radial distributions from the same projection head; Section 5.6 reports how much this matters.

4.2 Geometry-Specific Contrastive Similarity

The proposed margin acts on similarity logits and is independent of the metric used to produce them. The Euclidean control uses $\sigma(z_a, z_t) = z_a^\top z_t$; the hyperbolic variants use $\sigma(z_a, z_t) = -d_L(z_a, z_t)$. Writing σ rather than s avoids conflict with the caption string. Given a batch of N audio-text pairs $\{(z_{a,i}, z_{t,i})\}_{i=1}^N$, the base hyperbolic contrastive loss is:

$$\mathcal{L}_{\text{con}} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{e^{\sigma(z_{a,i}, z_{t,i})/\tau}}{\sum_j e^{\sigma(z_{a,i}, z_{t,j})/\tau}} + \log \frac{e^{\sigma(z_{t,i}, z_{a,i})/\tau}}{\sum_j e^{\sigma(z_{t,i}, z_{a,j})/\tau}} \right], \quad (6)$$

where $\tau > 0$ is a learnable temperature, initialised to 0.07. For the hyperbolic variants, the unbounded geodesic distance permits

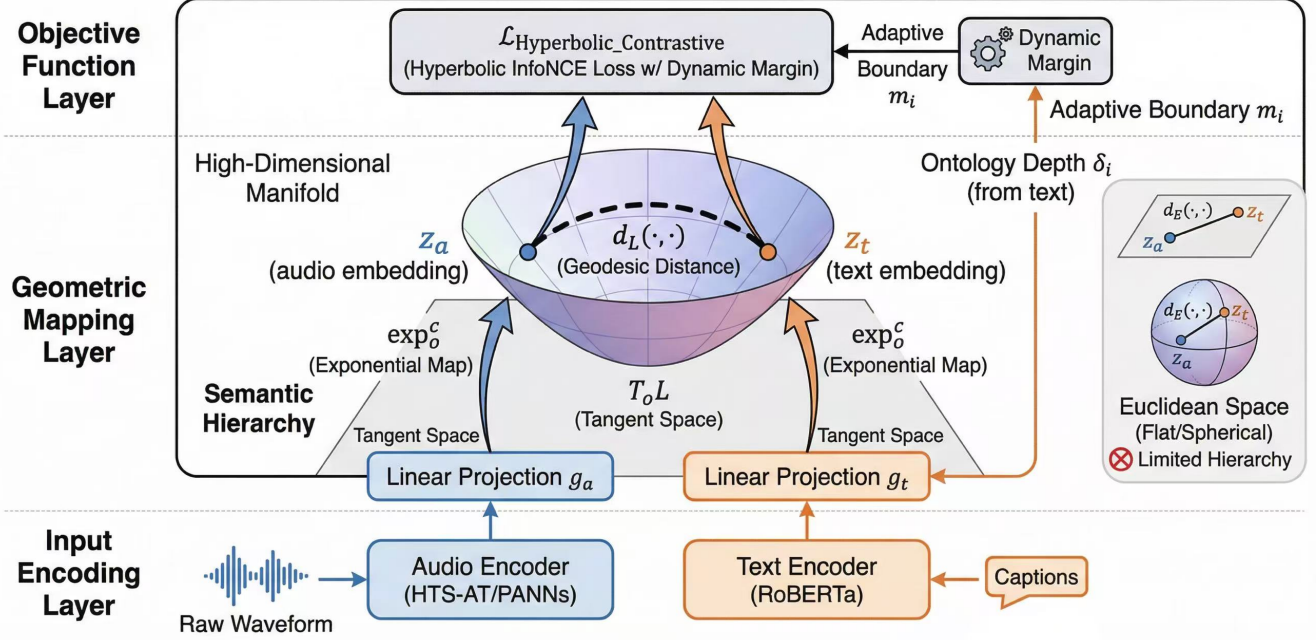


Figure 2: Dynamic Margin CLAP in its hyperbolic instantiation. Frozen HTS-AT and RoBERTa encoders feed modality-specific projections and a Lorentz lift. The proposed margin $m_i = m_0 + \alpha \delta_i$ modifies only matched logits; the same objective is applied to the Euclidean control and removed at inference.

unmatched pairs to disperse radially while matched pairs remain close. This no-margin objective isolates the effect of geometry; the proposed intervention in the next section changes only the positive logits and can be used with either similarity definition.

4.3 Dynamic Positive-Pair Margin

Standard CLAP assigns every matched pair the same positive target, even though caption specificity varies substantially. We replace this uniform treatment with a pair-dependent margin. Under hyperbolic similarity the same mechanism also limits excessive compression of matched pairs near the origin when distant negatives dominate the gradient. We estimate the caption specificity score $\delta_i \in [0, 1]$ as the normalised mean IDF of its tokens, computed over the training-split vocabulary after lowercasing and stopword removal:

$$\delta_i = \frac{1}{|W_i|} \sum_{w \in W_i} \frac{\text{idf}(w)}{\max_{w'} \text{idf}(w')}. \quad (7)$$

Here W_i denotes the set of unique non-stopword tokens in caption i , and IDF is computed on the training split of the corpus in use [17], so that no evaluation caption contributes to the statistic. Rare tokens tend to co-occur with semantically specific sound events (e.g., instrument names, onomatopoeia), making normalised IDF a lightweight, corpus-derived proxy for caption specificity—without requiring ontology access, POS tagging, or lemmatisation. The per-pair margin $m_i = m_0 + \alpha \cdot \delta_i$ is then subtracted from each

positive-pair similarity prior to softmax:

$$\mathcal{L}_{\text{mar}} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{e^{(\sigma(z_{a,i}, z_{t,i}) - m_i)/\tau}}{\sum_j e^{\sigma(z_{a,i}, z_{t,j})/\tau}} + \log \frac{e^{(\sigma(z_{t,i}, z_{a,i}) - m_i)/\tau}}{\sum_j e^{\sigma(z_{t,i}, z_{a,j})/\tau}} \right], \quad (8)$$

where m_0 is a base margin and α a scaling factor, following the additive-margin principle of ArcFace [5] and CosFace [34]. Semantically specific pairs (high δ_i) face a harder classification target and must achieve a larger positive–negative similarity gap; coarse-grained pairs (low δ_i) receive a smaller margin, relaxing constraints in ambiguous regions. At inference, the margin is removed and retrieval uses the native similarity of the selected geometry: dot product for Euclidean models or negative geodesic distance for hyperbolic models.

4.4 Training and Retrieval Procedure

The complete procedure has three stages. First, caption statistics are computed once from the active training split. We lowercase each caption, remove stopwords, retain unique content tokens, and cache the resulting specificity score δ_i and margin m_i with the training pair. Statistics are corpus-specific: WavCaps values are not reused for Clotho or MACS, and evaluation captions never contribute to IDF. This preprocessing introduces neither an auxiliary language model nor additional supervision.

Second, every mini-batch is passed through the frozen audio and text encoders and the two trainable projection heads. The projected

vectors are either normalised in Euclidean space or lifted with DirectLift or ExpMap. We then form all cross-modal similarities and apply the margin only to the matched logits in both retrieval directions. The negative set and its sampling distribution are unchanged. This distinction is important: loss re-weighting rescales the contribution of a complete example after its contrastive probabilities have been formed, whereas the margin directly changes the similarity gap required for the matched pair.

Third, retrieval discards the training-only margin. Queries and candidates are projected into the selected geometry, scored by dot product in the Euclidean control or negative geodesic distance in the hyperbolic variants, and ranked directly. Caption statistics and calibration are therefore unnecessary at test time. For N projected embeddings of dimension n , forming the full batch score matrix costs $O(N^2n)$ in either geometry. Scoring one query against a bank of B candidates costs $O(Bn)$ and can be chunked without changing the ranking. Lorentz similarity adds a constant-factor cost, which we measure in Section 5.8, but the dynamic margin does not change inference complexity.

Design properties. The method is intentionally modular. It does not modify either encoder, requires no ontology or task label, and can be attached to any similarity function that produces a matched audio–text logit. Its corpus-derived schedule is deterministic and auditable, while its training-only use preserves the original retrieval interface. These properties let the experiments separate three questions cleanly: whether a positive-pair margin helps at all, whether caption specificity improves over a uniform schedule, and whether curvature provides an additional representational benefit.

5 Experiments

5.1 Experimental Setup

What is trained. All reported experiments use frozen LAION-CLAP embeddings. The backbone comprises an HTS-AT audio encoder [2] and a RoBERTa text encoder [20]. Neither encoder is updated; only the two projection heads g_a and g_t are trained. Every corpus is trained and evaluated on its own split; we report no cross-corpus transfer. This design isolates the contribution of the positive-pair margin and the optional embedding geometry from any change in encoder capacity.

Datasets and evaluation protocol. We evaluate on the three corpora and splits of Table 1. The WavCaps–FreeSound results are obtained on a reconstructed 100K subset with a 70,000/30,000 split; they are not a reproduction of the split used in the original WavCaps paper and should not be compared against numbers reported under that protocol. MACS uses a deterministic audio-grouped 70/30 split (grouping seed 42), so that no audio clip contributes to both sides. The primary metric is Mean Recall (MR), the mean of Recall@{1, 5, 10} in the text-to-audio (T2A) and audio-to-text (A2T) directions. Retrieval uses chunked scoring throughout: the query–candidate score matrix is evaluated in blocks of 512 candidates so that peak memory stays bounded on the 30K WavCaps pool. Chunking changes only how the score matrix is computed, not the ranking or the metric. Because the three corpora have evaluation pools of very different sizes, absolute MR is not comparable across corpora; all comparisons below are within a corpus.

Table 1: Protocol for every reported result. IDF is computed only from the active training captions of the same corpus. Retrieval uses chunk size 512 throughout. WavCaps–FreeSound is a reconstructed 100K subset, not the split of the original WavCaps paper; its 30K pool doubles as the candidate bank for the scale and cost measurements of Section 5.8; the long-tail subsets used in the supplementary material are drawn from the same corpus.

Corpus	Split protocol	Train	Pool	Role
WavCaps–FreeSound	Reconstructed 100K, 70/30	70,000	30,000	main + cost
Clotho	Official dev → train, val → eval	19,195	5,225	main
MACS [22]	Audio-grouped 70/30, seed 42	12,099	5,175	main

Conditions. We factorise the proposed margin against geometry, giving six conditions. **Euclidean** uses flat-space dot-product similarity. **DirectLift** and **ExpMap** are the two Lorentz parameterizations in Section 4.1 and use negative geodesic distance. Each geometry is evaluated with and without the dynamic margin. The suffix **+DynMargin** denotes our proposed method; the DirectLift variant is the reference hyperbolic instantiation. All three margin variants test whether the method transfers across geometry choices. ExpMap remains in every table so that parameterization sensitivity can be read off directly, and Section 5.6 states where it is the better lift.

Implementation. Training uses AdamW [21] (lr 3×10^{-4} , weight decay 10^{-4}) with cosine decay, batch size 256, for 40 epochs; the full configuration is given in the supplementary material. The margin schedule ($m_0 = 0.05$, $\alpha = 0.5$) and the geometry ($c = 1.5$, projection dimension 768, residual weight 0.3) were fixed once on WavCaps (Table 6) and reused unchanged on Clotho and MACS; we did not tune per corpus. All results are averaged over the same five seeds {42, 123, 3407, 777, 2023}; we report mean $\pm$ std over seeds. All timings and memory figures are measured on one NVIDIA GeForce RTX 4090.

5.2 Main Results

Table 2 and Figure 3 show two clear patterns. First, the dynamic margin improves mean recall in every geometry–corpus combination and is removed at inference. Second, the benefit of curvature is corpus-dependent. On WavCaps, both hyperbolic margin variants exceed the best Euclidean configuration beyond the seed spread. On Clotho and MACS, the best hyperbolic and Euclidean margin variants are effectively tied, and DirectLift without the margin can underperform the Euclidean baseline. We therefore claim a consistent margin effect, but not a universal curvature advantage.

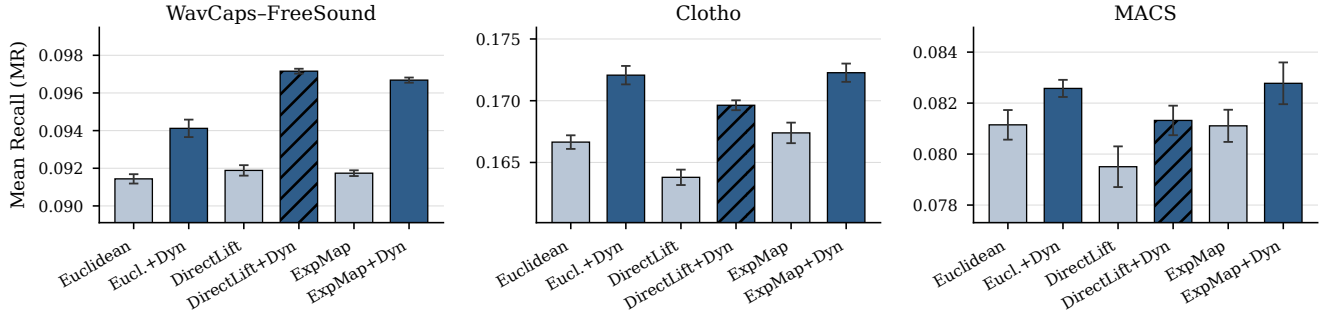
Parameterization also matters: DirectLift is strongest on WavCaps, whereas ExpMap is ahead on Clotho and MACS with all five seeds agreeing. The hyperbolic lift should thus be treated as a corpus-dependent design choice rather than an implementation detail.

5.3 Effect of the Dynamic Margin

Table 3 compares each geometry with its own no-margin counterpart. The primary observation is the consistency of the effect:

Table 2: Cross-modal retrieval (mean $\pm$ std over five seeds). MR averages R@1/5/10 in both directions. Bold is best per corpus; each +DynMargin row uses the proposed objective. Scores are comparable only within a corpus.

Corpus	Condition	T2A			A2T			MR
		R@1	R@5	R@10	R@1	R@5	R@10	
WavCaps–FreeSound	Euclidean	0.0266	0.0926	0.1458	0.0297	0.0996	0.1544	0.091437 $\pm$ 0.000248
	Eucl.+DynMargin	0.0276	0.0945	0.1490	0.0314	0.1039	0.1584	0.094120 $\pm$ 0.000464
	DirectLift	0.0274	0.0927	0.1463	0.0295	0.0999	0.1557	0.091886 $\pm$ 0.000278
	DirectLift+DynMargin	0.0290	0.0985	0.1529	0.0327	0.1069	0.1630	0.097157$\pm$0.000129
	ExpMap	0.0276	0.0928	0.1451	0.0300	0.0998	0.1552	0.091739 $\pm$ 0.000152
	ExpMap+DynMargin	0.0291	0.0978	0.1525	0.0321	0.1063	0.1623	0.096682 $\pm$ 0.000137
Clotho	Euclidean	0.0402	0.1818	0.2727	0.0469	0.1797	0.2787	0.166648 $\pm$ 0.000551
	Eucl.+DynMargin	0.0423	0.1892	0.2842	0.0493	0.1851	0.2823	0.172070 $\pm$ 0.000747
	DirectLift	0.0375	0.1736	0.2705	0.0480	0.1777	0.2755	0.163796 $\pm$ 0.000624
	DirectLift+DynMargin	0.0404	0.1828	0.2790	0.0490	0.1830	0.2836	0.169633 $\pm$ 0.000400
	ExpMap	0.0397	0.1801	0.2769	0.0473	0.1824	0.2779	0.167394 $\pm$ 0.000832
	ExpMap+DynMargin	0.0423	0.1871	0.2849	0.0489	0.1859	0.2846	0.172268$\pm$0.000740
MACS	Euclidean	0.0192	0.0820	0.1321	0.0259	0.0889	0.1387	0.081147 $\pm$ 0.000581
	Eucl.+DynMargin	0.0191	0.0849	0.1334	0.0268	0.0904	0.1408	0.082576 $\pm$ 0.000336
	DirectLift	0.0190	0.0805	0.1286	0.0251	0.0874	0.1364	0.079504 $\pm$ 0.000798
	DirectLift+DynMargin	0.0183	0.0831	0.1329	0.0251	0.0889	0.1396	0.081320 $\pm$ 0.000579
	ExpMap	0.0198	0.0827	0.1319	0.0264	0.0878	0.1380	0.081108 $\pm$ 0.000631
	ExpMap+DynMargin	0.0197	0.0837	0.1343	0.0274	0.0912	0.1403	0.082776$\pm$0.000819

**Figure 3: Mean Recall for all six conditions (mean $\pm$ std over five seeds). Dark bars use the dynamic margin; light bars do not. The hatched bar is DirectLift+DynMargin. Axes are truncated and corpus-specific. Within each geometry, the margin improves all three corpora.**

introducing the proposed positive-pair margin benefits every evaluated corpus and every geometry. Adaptive optimisation pressure is therefore not tied to a particular embedding space and constitutes a robust improvement to contrastive audio–language training.

The interaction with geometry is nevertheless informative. On WavCaps, the largest and most heterogeneous corpus, both hyperbolic variants obtain larger improvements than the Euclidean control. This pattern suggests that additional representational capacity can reinforce the optimisation benefit introduced by the margin when the corpus contains richer semantic variation. On Clotho and MACS, however, the differences between geometries

are much smaller, indicating that the margin itself rather than the embedding manifold is the dominant source of improvement. Geometry therefore modulates the magnitude of the gain instead of determining whether the gain exists.

These observations support the central design principle of Dynamic Margin CLAP. The same objective transfers across similarity functions without changing the encoders, negative sampling, or retrieval pipeline, so the margin is best viewed as an independent training component. With five paired seeds, the smallest attainable one-sided p -value is 0.031; we therefore interpret the uniformly positive differences as a consistent directional result rather than a

Table 3: Margin effect Δ MR relative to each geometry’s no-margin control. All nine differences are positive.

Geometry	WavCaps	Clotho	MACS
Euclidean	+0.002683	+0.005422	+0.001429
DirectLift	+0.005271	+0.005837	+0.001816
ExpMap	+0.004943	+0.004874	+0.001668

broad significance claim. Overall, the experiments separate a general optimisation effect from a conditional representational effect: the margin remains beneficial in every setting, while curvature changes only how strongly that benefit is expressed on a particular corpus.

5.4 Comparison with Long-Tail Re-weighting

If the margin merely up-weighted rare captions, a standard long-tail re-weighting scheme applied to the same objective should reproduce its effect. Table 4 shows that it does not. On WavCaps under DirectLift, IDF re-weighting of the contrastive loss recovers 0.000448 of the 0.005271 that the margin obtains, and focal-style contrastive weighting is substantially worse than applying nothing at all (-0.022055 relative to the no-margin control). The two mechanisms are therefore not interchangeable: the margin modifies the positive-pair term inside the softmax, whereas re-weighting rescales whole examples. These two controls are not an exhaustive survey of long-tail learning: re-sampling [12] manipulates the training distribution rather than the objective, and label-hierarchy regularisation [15] requires the ontology our setting is designed to do without, so neither is a matched substitute for a per-pair margin. We give aggregate MR only and make no claim about which subgroups of queries the gain comes from (Section 5.9).

Table 4: Long-tail re-weighting baselines on WavCaps under DirectLift (mean $\pm$ std over 5 seeds). Δ is relative to the no-margin control in the first row.

Objective	MR	Δ
No margin (control)	0.091886 $\pm$ 0.000278	—
IDF re-weighting	0.092334 $\pm$ 0.000142	+0.000448
Focal-style weighting	0.069831 $\pm$ 0.000176	-0.022055
Dynamic margin	0.097157$\pm$0.000129	+0.005271

5.5 Multi-Positive Evaluation

Clotho pairs several captions with each clip, so a margin that only helped under one-to-one supervision would be of limited use. Table 5 re-scores Clotho with audio-id-level MR, evaluating retrieval per clip rather than per caption. The margin improves all three geometries again (+0.0058, +0.0067 and +0.0070), reproducing Table 3 under a different aggregation. Curvature does not benefit from the change: Eucl.+DynMargin is nominally the best condition here, with ExpMap+DynMargin within one standard deviation and DirectLift+DynMargin 0.0029 behind. The supplementary material

reports the same six conditions on frequency-stratified long-tail subsets of WavCaps, with the same ordering.

Table 5: Clotho multi-positive evaluation: audio-id-level MR (mean $\pm$ std over the same 5 seeds and protocol as Table 2).

Condition	Audio-id MR
Euclidean	0.332089 $\pm$ 0.002500
Eucl.+DynMargin	0.337907$\pm$0.001701
DirectLift	0.328351 $\pm$ 0.001561
DirectLift+DynMargin	0.335005 $\pm$ 0.001197
ExpMap	0.330699 $\pm$ 0.001479
ExpMap+DynMargin	0.337665 $\pm$ 0.001671

5.6 Which Parameterization?

The two Lorentz lifts are not interchangeable, and no single corpus settles the choice. A paired per-seed comparison places ExpMap ahead on Clotho, MACS, and Clotho multi-positive evaluation on all five seeds. On WavCaps the sign reverses, although the gap between the two lifts is small. The parameterisation therefore changes the radial distribution induced by the same projection head in a corpus-dependent way.

We retain DirectLift as the reference hyperbolic instantiation because it is cheaper to train and is strongest on the largest corpus. ExpMap, however, is preferable on the two smaller corpora and in the multi-positive protocol. This result reinforces the paper’s broader separation: the margin is stable across geometry choices, whereas the benefit of a particular hyperbolic mapping depends on corpus scale, semantic structure, and the way radii are allocated. In practice, the lift should be treated as a model-selection choice rather than as a fixed component of the proposed objective.

5.7 What the Margin Needs

Table 6 separates the benefit of introducing a positive-pair margin from the benefit of the proposed IDF schedule. Mean IDF is the strongest setting under both hyperbolic lifts, and averaging token rarity across a caption is more reliable than anchoring the schedule to only its rarest token. The selected schedule is therefore a sensible corpus-derived instantiation of the general objective.

The broader result is more important: the margin source explains only a small fraction of the total improvement. Uniform and random margins recover roughly 95% of the mean-IDF gain under DirectLift. Enforcing an additional positive–negative similarity gap is thus substantially more influential than the exact statistic used to set that gap. We accordingly interpret IDF as a lightweight and systematic refinement, not as the sole mechanism behind the method.

Caption length provides a useful counterexample. Its performance stays close to the no-margin control, showing that merely varying the margin across examples is insufficient; the conditioning signal must preserve some relation to semantic specificity. The ablation therefore supports a three-level conclusion: positive-pair margining carries the main effect, token rarity supplies a smaller increment, and a weak proxy such as caption length does not reproduce that increment. Future schedules based on uncertainty or

Table 6: Margin ablation on WavCaps (MR, mean $\pm$ std over 5 seeds), for both parameterizations. Bold: best per column.

Margin source	DirectLift	ExpMap
None	0.091886 $\pm$ 0.000278	0.091739 $\pm$ 0.000152
Mean IDF (final)	0.097157$\pm$0.000129	0.096682$\pm$0.000137
Mean IDF (intermediate)	0.095914 $\pm$ 0.000252	0.095584 $\pm$ 0.000180
Max IDF	0.096767 $\pm$ 0.000041	0.096182 $\pm$ 0.000196
Uniform	0.096869 $\pm$ 0.000172	0.096440 $\pm$ 0.000130
Random	0.096848 $\pm$ 0.000132	0.096451 $\pm$ 0.000112
Caption length	0.092351 $\pm$ 0.000245	0.092148 $\pm$ 0.000228

Table 7: Candidate-bank scoring latency and peak GPU memory (RTX 4090, chunk size 512). Scoring only, not end-to-end ranking.

Candidates	Time (ms)			Peak mem. (MB)		
	Eucl.	DirectLift	ExpMap	Eucl.	DirectLift	ExpMap
1,000	0.302	0.464	0.501	405.9	407.7	414.0
10,000	0.600	0.911	0.904	467.7	510.6	498.3
30,000	1.510	3.731	3.541	601.7	737.0	686.3
100,000	4.680	12.460	11.947	1074.2	1532.1	1344.9

learned specificity should consequently be compared first against a uniform positive margin, not only against the no-margin baseline.

5.8 What Curvature Costs

Table 7 separates the deployment cost of the optional geometry from that of the proposed objective. The dynamic margin is removed after training and adds no retrieval-time latency, memory use, caption processing, or calibration stage. All measured inference overhead therefore comes from Lorentz rather than Euclidean similarity.

At a 100K candidate bank, DirectLift and ExpMap require 2.66 $\times$ and 2.55 $\times$ the Euclidean scoring time, respectively, with higher peak GPU memory. The gap is smaller for a 1K bank and increases with retrieval scale. Although both geometries retain the same asymptotic scoring complexity, the Lorentz inner product, geodesic distance, and inverse hyperbolic cosine introduce a non-negligible constant factor. The relevant deployment variable is therefore candidate-bank size, not geometry in isolation.

The margin and geometry can consequently be selected independently. Dynamic Margin CLAP can be added to a Euclidean retrieval system without changing its inference budget, whereas hyperbolic projection is justified only when its corpus-dependent benefit outweighs the extra scoring cost. In our setup, a WavCaps run takes 82.5 s for Euclidean projection, 101.9 s for DirectLift, and 159.2 s for ExpMap. This separation preserves the margin benefit in latency-sensitive systems while leaving curvature as an explicit deployment choice rather than a requirement of the training objective.

5.9 Discussion and Limitations

Primary finding and scope. Positive-pair margining is the only effect that persists across every corpus, geometry, and evaluation protocol. Curvature changes representational capacity, whereas the margin directly changes the training target; IDF is a lightweight

schedule rather than the sole source of improvement. This distinction matters for interpretation: the evidence supports adaptive positive-pair objectives broadly, but supports hyperbolic projection only conditionally.

Representation and data limits. All results use frozen encoders and trainable projection heads. This design isolates the objective and geometry, but cannot establish whether either remains the binding constraint under end-to-end training. The largest training split contains 70K pairs, and music, long-form audio, multilingual captions, and million-pair regimes remain untested. Audio semantics are also compositional and multi-relational, not a strict tree; Lorentz space should therefore be interpreted as an inductive bias for the approximately hierarchical, heavy-tailed part of the data rather than as a complete semantic model.

Margin limits. Uniform and random margins recover most of the gain, supporting the positive margin itself more strongly than the IDF proxy. We do not report per-query or frequency-stratified causal explanations for the aggregate improvement, nor do we compare uncertainty-, perplexity-, or model-derived specificity signals. More seeds, alternative backbones, convergence analysis, and end-to-end training would strengthen the statistical and architectural scope. These limitations bound the claim but also identify a clear next step: evaluate richer margin schedules against the strong uniform-margin control while preserving the geometry-independent protocol used here.

6 Conclusion

Dynamic Margin CLAP replaces uniform positive-pair pressure with a caption-conditioned margin in contrastive audio–language learning. Across the evaluated corpora, geometries, and multi-positive protocol, the margin provides the only consistently positive intervention and adds no inference-time component. The ablations further show that most of this benefit comes from introducing a positive margin itself, while normalised IDF supplies a smaller specificity-dependent refinement. Lorentz projection is complementary: it helps on WavCaps, remains approximately neutral on the smaller corpora, and incurs additional retrieval cost. The principal result is therefore the utility of adaptive positive-pair objectives, with geometry selected according to corpus structure and deployment constraints.

Ethical considerations. This study uses public English audio-caption corpora and released frozen encoders, with no human subjects or new data collection. The method can support accessible audio search and archival retrieval, but deployment for audio monitoring may raise privacy concerns. Its caption-derived margin also inherits corpus and vocabulary biases, so the reported gains do not establish equitable performance across speakers, languages, or acoustic environments.

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